

Part 1:

The screenshot shows the Databricks workspace interface for the workspace `istm637-lakehouse-537002266`. At the top, there's a navigation bar with tabs for Code, Issues, Pull requests, Agents, Actions, Projects, Security and quality, Insights, and Settings. Below this, the 'Commits' section is active, showing a list of recent commits on the `main` branch. The commits include 'Edit #2' (4a02225), 'Small edit' (86d227f), and 'Add files via upload' (60b6d7d). Below the commits, the file editor for `ISTM637_Databricks_Project_Starter.ipynb` is shown. The editor displays a diff view with changes to the notebook's code. The left sidebar shows the file explorer with the selected file. The bottom of the editor has a commit message field and a 'Commit & push' button.

Part 2:

```

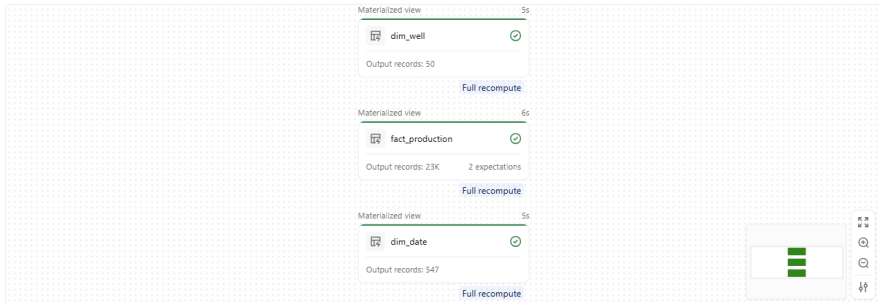
# Confirm the three files are in the volume
files = [f.name for f in dbutils.fs.ls(VOL_PATH)]
print("Files in volume:", files)
expected = {"dim_well.csv", "dim_date.csv", "fact_production.csv"}
missing = expected - set(files)
assert not missing, f"Still need to upload: {missing}"
print("All three CSVs found ✓")

```

Files in volume: ['dim_date.csv', 'dim_well.csv', 'fact_production.csv']
All three CSVs found ✓

New Pipeline 2026-07-29 16:52 ☆ Send feedback

Run: Jul 29, 2026, 05:02 PM · Completed · Select tables for refresh



Pipeline details

Pipeline ID: 0f60a58a-cdea-40c5-aed2-3420e8fc99ee
Pipeline type: ETL pipeline
Creator: emily.cendejas@tamu.edu
Owner: emily.cendejas@tamu.edu
Run as: emily.cendejas@tamu.edu
Source code: 1 folder · Open in Editor

Run details

Pipeline run ID: 8e566899-d720-4258-98fd-22333c3ce2f7
Run status: Completed
Run type: Refresh all · Learn more

Tables 3 Performance 3

Filter by table Type Status

Name	Duration	Output records	Expectations	Incrementalization
dim_date	5s	547	-	Full recompute
dim_well	5s	50	-	Full recompute
fact_production	6s	23K	2 met	Full recompute

Catalog Explorer > istm637_emily_cendejas > oilgas > raw

Overview Files Details Permissions Policies

Description Raw uploaded CSVs

About this volume Owner: emily.cendejas@tamu.edu

Tags Add tags

/Volumes/istm637_emily_cendejas/oilgas/raw

Filter files and directories at this level

Name	Size	Last modified
dim_date.csv	38.23 KB	21 minutes ago
dim_well.csv	7.51 KB	21 minutes ago
fact_production.csv	1.26 MB	21 minutes ago

databricks

Catalog Explorer > istm637_emily_cendejas > oilgas

Overview Details Permissions Policies

Description Oil & gas star schema

About this schema Owner: emily.cendejas@tamu.edu

Tags Add tags

Policies New policy

Filter tables

Name	Owner	Created at	Popularity	Comment
dim_date	emily.cendejas@tamu.e...	Jul 29, 2026, 05:03 PM	---	Calendar date dimension...
dim_well	emily.cendejas@tamu.e...	Jul 29, 2026, 05:03 PM	---	Well master data: one r...
fact_production	emily.cendejas@tamu.e...	Jul 29, 2026, 05:03 PM	---	Daily production fact ta...

Part 3:

data

bricks

New

Home

Learn

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Discover

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie Agents

Alerts

Query History

SQL Warehouses

Data Engineering

Runs

Data Ingestion

Visual Data Prep

AI/ML

Playground

Agents

AI Gateway

Catalog

Serverless Starter Warehouse Serverless

Catalog

Type to search

My organization

workspace

system

istm637_emily_cendejas

default

information_schema

oilgas

(3)

dim_date

dim_well

fact_production

Volumes (1)

raw

Shares received

samples

Search data, notebooks, recents, and more...

CTRL + P

dim_well

Overview

Sample Data

Details

Permissions

Policies

History

Lineage

Insights

Quality

Description

Well Master Data Table Description

This table captures essential information for each well, including:

Attributes

Location: Details about where each well is situated.

Show more

Definition

SQL

```

1 CREATE MATERIALIZED VIEW 'istm637_emily_cendejas'.oilgas.'dim_well' (
2   well_id STRING COLLATE UTF8_BINARY
3   COMMENT 'A unique identifier assigned to each well, facilitating tracking and reference across different datasets.',
4   well_name STRING COLLATE UTF8_BINARY
5   COMMENT 'The official name of the well, which is often used to identify it in reports and discussions.',
6 )

```

Filter columns...

Column	Type	Popularity	Comment	Tags	Column masking
well_id	string	---	A unique identifier assigned to each well, facilitating tracking and reference across different datasets.		
well_name	string	---	The official name of the well, which is often used to identify it in reports and discussions.		

About this materialized view

Owner: emily.cendejas@tamuedu

Popularity: ---

Current pipeline update status

Completed Jul 29, 2026, 09:03 PM (22 h ago)

See refresh details

Refresh schedule

No refresh schedules

Quality

Not enabled

Data Quality Monitoring is not enabled for this table

Enable

Policies

New policy

Tags

domain:production layer:silver pii:none

Row filter

Add filter

Insights

The screenshot displays the Databricks Catalog sidebar on the left, which lists various data sources like SQL Editor, Queries, Dashboards, Genie Agents, Alerts, Query History, and SQL Warehouses. The main pane shows the 'Catalog' tree under 'Serverless Starter Warehouse'. It details a hierarchy from 'My organization' down to 'fact_production' table and 'raw' volume. A search bar at the top allows filtering by name or path.

Catalog

Serverless Starter Warehouse Serverless 2XS

Catalog

Type to search

My organization

workspace

system

istm637_emily_cendejas

default

information_schema

oilgas

Tables (3)

dim_date

dim_well

fact_production

Volumes (1)

raw

Shares received

samples

Catalog Explorer > istm637_emily_cendejas > oilgas >

fact_production

Overview Sample Data Details Permissions Policies History Lineage Insights Quality

Description

Daily production fact table: one row per well per producing day. Grain = well_id + date_id. Ingested from raw CSV via Lakeflow.

Definition SQL

```
1 CREATE MATERIALIZED VIEW 'istm637_emily_cendejas'.fact_production` (  
2   production_id INT  
3   COMMENT 'A unique identifier for each production record, facilitating tracking and reference for specific entries in the table.'  
4   well_id STRING COLLATE UTF8_BINARY  
5   COMMENT 'Identifier representing the specific well from which the production data is sourced, linking the record to a particular extraction site.'  
... 30 more lines
```

Filter columns...

Column	Type	Popularity	Comment	Tags	Column masking
production_id	int	---	A unique identifier for each production record, facilitating tracking and reference for specific entries in the table.		
well_id	string	---	Identifier representing the specific well from which the production data is sourced, linking the record to a particular extraction site.		
date_id	int	---	Represents the date of the production data, allowing for chronological tracking.		

About this materialized view

Owner emily.cendejas@tamu.edu

Popularity

Current pipeline update status

Completed Jul 29, 2026, 05:03 PM (22 h ago)

See refresh details

Refresh schedule

No refresh schedules

Quality

Not enabled

Data Quality Monitoring is not enabled for this table

Enable

Policies

New policy

Tags

domain:production layer:silver pii:none

Row filter

Add filter

Insights

No active user in past 30 days

Catalog

Serverless Starter Warehouse Serverless 2XS

Catalog

Type to search

My organization

workspace

system

istm637_emily_cendejas

default

information_schema

oilgas

Tables (3)

dim_date

dim_well

fact_production

Volumes (1)

raw

Shares received

samples

oil_bbl

double

Volume of oil produced on the given date, measured in barrels (bbl). A barrel is 42 U.S. gallons.

gas_mcf

double

Volume of gas produced on the given date, measured in thousand cubic feet (Mcf)

water_bbl

double

Volume of water produced on the given date, measured in barrels (bbl).

downtime_hours

double

The total hours during which the well was not operational on the specified date, useful for analyzing production efficiency.

choke_size_64ths

int

The size of the choke used in the well, measured in 64ths of an inch, which can affect production rates and pressures.

tubing_pressure_psi

int

The pressure measured in pounds per square inch within the tubing of the well, critical for assessing operational conditions.

casing_pressure_psi

int

The pressure measured in pounds per square inch within the casing of the well, important for monitoring well

Part 4:

Which basin produced the most oil overall? Reviewable

Chat Monitor Benchmark

List the top 5 wells by total gas production.

> Thinking complete

Top 5 Wells by Total Gas Production

The top gas-producing well is Weld 35-34 in the DJ Basin, operated by EOG Resources, with 1.13 million Mcf of total gas production ¹.

[Show code](#)

Top 5 Wells by Total Gas Production			
Well Name	Operator	Basin	Total Gas Production
Weld 35-34	EOG Resources	DJ Basin	1133126 Mcf
Lea 20-2	Occidental	Delaware Basin	1020380 Mcf
Martin 28-2H	Coterra	Permian Basin	955693 Mcf
La Salle 5-11H	Occidental	Eagle Ford	933023 Mcf
Eddy 25-15H	ConocoPhillips	Delaware Basin	910567 Mcf

Source:  Top 5 wells by total gas production

The top five wells span multiple basins and operators. Occidental operates two wells in the top five: Lea 20-2 in the Delaware Basin (1.02 million Mcf) and La Salle 5-11H in Eagle Ford (933K Mcf). Coterra's Martin 28-2H in the Permian Basin ranks third with 956K Mcf, while ConocoPhillips' Eddy 25-15H in the Delaware Basin rounds out the top five with 911K Mcf ¹.

> Encoded HTML

Ask your question...



← List the top 5 wells by total gas production. 

```
1 SELECT
2   f.well_id,
3   w.well_name,
4   ROUND(SUM(f.gas_mcf)) AS total_gas_mcf
5 FROM istm637_emily_cendejas.oilgas.fact_production AS f
6 JOIN istm637_emily_cendejas.oilgas.dim_well AS w
7   ON f.well_id = w.well_id
8 GROUP BY
9   f.well_id,
10  w.well_name
11 ORDER BY total_gas_mcf DESC
12 LIMIT 5;
```

Part 5:

Run37 minutes agoDefault (workspace)Default (default)Edit

```
1 SELECT
2   f.production_id,
3   f.well_id,
4   f.date_id,
5   f.oil_bbl,
6   f.gas_mcf,
7   f.water_bbl,
8   f.downtime_hours,
9   f.choke_size_64ths,
10  f.tubing_pressure_psi,
11  f.casing_pressure_psi,
12  w.well_name,
13  w.operator,
14  w.field_name,
15  w.basin.
```

+ Add parameter

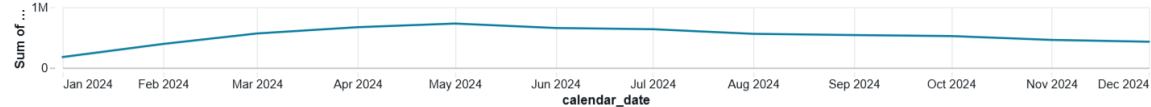
Result TableSchema+ Add custom calculation

	1.2 production_id	1.2 well_id	1.2 date_id	1.2 oil_bbl	1.2 gas_mcf	1.2 water_bbl	1.2 downtime_hours	1.2 choke_s
1	1	WELL-0001	20240513	1188	3399.6	1057.1	0	
2	2	WELL-0001	20240514	1227.3	3455.3	1094.8	0	

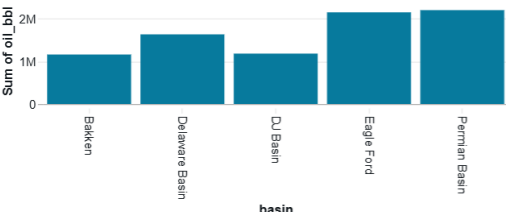
See performance (1)Checked for improvements

New Dashboard2026-07-30 19:12:24☆Edit draft34m agoScheduleShare

Monthly Oil Production Trend (2024)
Shows total oil production over time.



Oil Production by Basin
Compares total oil production across basins.



Total Oil Produced
Total oil production across the filtered data.

8.32M

Total Gas Produced
Total gas production across the filtered data.

21.62M

Filter by Basin

All

Ask Genie

Part 6:

4 — Evaluate

▶ 2 minutes ago (1s) 9

```
import numpy as np
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

pred = model.predict(X_test)
mae = mean_absolute_error(y_test, pred)
rmse = mean_squared_error(y_test, pred) ** 0.5
r2 = r2_score(y_test, pred)
print(f"MAE = {mae:,.1f} bbl/day")
print(f"RMSE = {rmse:,.1f} bbl/day")
print(f"R2 = {r2:,.3f}")
# TODO (extend): try adding interaction features or a different regressor and beat this R2.
```

MAE = 33.5 bbl/day
RMSE = 70.0 bbl/day
R2 = 0.933

Catalog Explorer > istm637_emily_cendejas >

oilgas

⋮

Use with BI tools

Share

Create

Overview Details Permissions Policies

Description

Oil & gas star schema

Tables 4 Volumes 1 Models 1 Functions 1 Secrets 0 Services 0 Connections 0

Sort ▾

Name	Owner	Created at
oil_rate_predictor	emily.cendejas@tamu.edu	Jul 30, 2026, 08:08 PM

About this schema

Owner emily.cendejas@tamu.edu

Catalog Explorer > istm637_emily_cendejas > oilgas >

oil_rate_predictor

⋮

Share

Serve this model

Overview Details Permissions Policies Lineage

Description

AI generate Add

Versions

Status	Version	Tags	Aliases	Deployment jobs	Active endpoints	Comment
	Version 1		@ champion			

Activity log

Activity logs capture updates from deployment jobs. [Learn more.](#)

Time	Version	Log
2026-07-30 20:08:16	Version 1	Model version 1 registered by user emily.cendejas@tamu.edu.

About this model

Owner emily.cendejas@tamu.edu

ISTM637_Databricks_Project_Starter																																																																																				
ISTM637_Predictive_Model_Notebook																																																																																				
New Query 2026-07-30 20:25:49																																																																																				
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<pre> 1 SELECT * 2 FROM istm637_emily_cendejas.oilgas.well_forecast 3 WHERE well_id = 'WELL-0001' </pre>																																																																																				
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180 rows 0.645s runtime See performance (1) Checked for improvements																																																																																				

Part 7:

Warehouse ID: 1f991176c65beb72

Host: https://dbc-09ec943c-37a9.cloud.databricks.com

Oil & Gas Well Production Analytics

ISTM 637 Project - Emily Cendejas

Select a Well

Choose a wellid:

WELL-0001

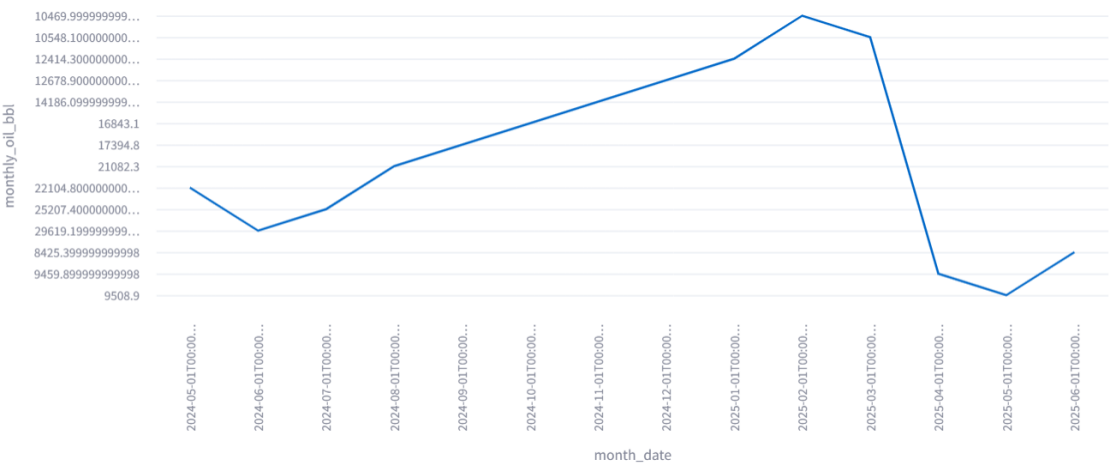
Selected Well: WELL-0001

Historical Monthly Oil Production

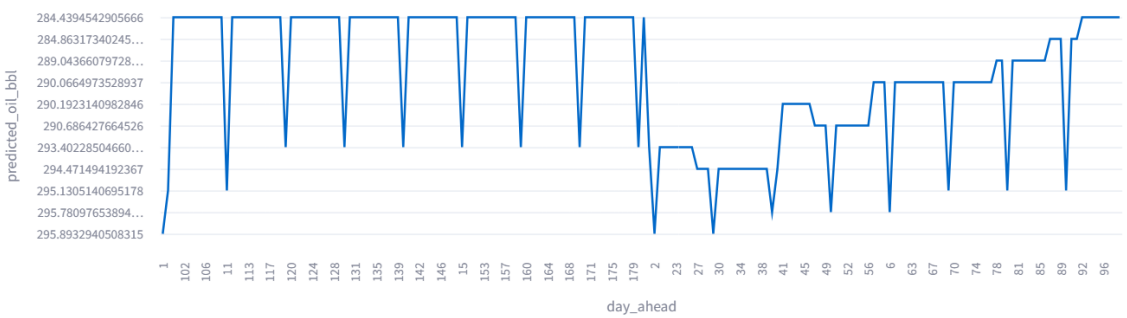
thly_oil_bbl

⋮

Historical Monthly Oil Production



Forecasted Daily Oil Production



Step 8:

Share data

- 1 Create share
- 2 Add data assets
- 3 Add notebooks
- 4 Add recipients

Select assets across catalogs and schemas

Type to search...

For you

All

istm637_emily_cendejas

default

oilgas

Tables (4)

dim_date

☒ dim_well

fact_production

well_forecast

Volumes (1)

workspace

Type	Name	Alias	History	Partition
	istm637_emily_cendejas...		-	-

Tables are shared with full history and commit log by default. You can choose to exclude sharing history for one or more tables.

Share data

- 1 Create share
- 2 Add data assets
- 3 Add notebooks
- 4 Add recipients

Recipient	Authentication type	Description
benitaanah	DATABRICKS us-east-2	
Select a recipient		

CancelBackShare data

dim_well_share

Overview Recipients

Filter recipients1 recipient

Status	Name	Authentication type	Created at	Owner	Description
	benitaanah	DATABRICKS us-east-2	Jul 30, 2026, 10:08 PM	emily.cendejas@tamu.edu	